Combinatorial proof of Pascal's identity

Suppose S is a set containing n elements. Let a be an element of S and $T = S - \{a\}$.

There are $\binom{n}{k}$ subsets of S containing k elements.

However, a subset of S with k elements either contains a together with $k-1$ elements of T, or contains k elements of T and does not contain a.

Because there are $\binom{n-1}{k-1}$ subsets of $k-1$ elements of T, there are $\binom{n-1}{k-1}$ subsets of k elements of S that contain a.

Because there are $\binom{n-1}{k}$ subsets of k elements of T , there are $\binom{n-1}{k}$ subsets of k elements of S that don't contain a.

Therefore: $$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$